

Chapter 33: Complications of the Term Newborn —

Revision Notes

Williams Obstetrics 26e, pp. 1545–1587 of the PDF. High-yield numbers are in **bold**. The four tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: Many neonatal problems are extensions of fetal pathology. **Most cerebral palsy (CP) begins long before labor:** CP rates have stayed flat for 50 years despite a **sixfold rise in cesarean rates**. Only a small fraction of CP is due to intrapartum hypoxia.

1. Respiratory Distress

- Signs: hypoxemia with compensatory **tachypnea, nasal flaring, chest wall retractions, grunting**.
- Preterm: mostly **RDS** (surfactant deficiency, Ch 34).
- **Leading causes at term: transient tachypnea of the newborn (TTN), meconium aspiration syndrome, pneumonia, pneumothorax.**
 - Less common: RDS, persistent pulmonary hypertension, acid–base disturbance, CNS insults, congenital chest/airway/CNS/cardiac anomalies.
- **Evaluation:** chest radiograph; CBC or CRP and blood cultures (infection); arterial blood gas.
- **Treatment is supportive:** oxygen as needed → CPAP or mechanical ventilation. Feeding by mouth, gavage or IV depending on the degree of tachypnea (high rates limit breastfeeding and raise aspiration risk).

Transient tachypnea of the newborn

- Benign, self-limited; due to **slow clearance of fetal lung fluid**.
- Risk factors: delivery **<39 wk** (risk rises as GA falls), **cesarean without labor**, male sex, perinatal asphyxia, LGA or SGA, maternal GDM or asthma.
- **Diagnosis of exclusion.** Supportive O₂. Most resolve **within 48 h**.

Respiratory distress syndrome at term

- Surfactant deficiency; low incidence at term.
- Risks: **male sex, white race, maternal diabetes**. Rare surfactant gene mutations: **SP-B, SP-C, ABCA3, thyroid transcription factor 1**.
- Treatment as for preterm: CPAP/ventilation ± surfactant (ET tube or less invasive techniques).
- **GBS pneumonia is often indistinguishable from RDS** clinically and radiographically → empirical antibiotics are often added.
- **Late-preterm (34–37 wk) antenatal corticosteroids** enhance surfactant synthesis. Parkland does **not** give them for this indication: concern for **neonatal hypoglycemia** and unknown long-term effects (promptly treated hypoglycemia seems to leave no sequelae).

Meconium aspiration syndrome

- Mechanisms: acute **airway obstruction, chemical pneumonitis, surfactant dysfunction/inactivation, pulmonary hypertension**.
- Meconium-stained fluid is common; severe aspiration is not: **~4 per 1000** newborn discharges, rising from **37 to 43 wk**.
- Morbidity is linked to **thick meconium**, but MAS can follow light staining and a normal labor.

- **Highest risk: postterm pregnancy and FGR** — less amniotic fluid, cord compression or uteroplacental insufficiency → thick, undiluted meconium.

Prevention — what does NOT work

Intervention	Evidence
FHR abnormalities to predict aspiration	Unreliable predictor
Oropharyngeal suctioning at the perineum	Abandoned: no ↓ in MAS incidence or severity
Routine endotracheal intubation + suction (vigorous or nonvigorous)	No benefit; no longer recommended routinely
Amnioinfusion for thick meconium (Fraser 2005, ~2000 women ≥36 wk)	Perinatal death 0.05% in both groups ; mod–severe MAS 4.4% vs 3.1% (NS); cesarean 32% vs 29%

- Pediatric providers must still attend deliveries: these neonates may need **effective positive-pressure ventilation**.
- Amnioinfusion still has a role for **recurrent variable decelerations** with low fluid (Ch 24), not for meconium.

Treatment

- Ventilatory support (CPAP, intubation, mechanical ventilation).
- **Surfactant replacement**: may ↓ need for **ECMO**, but does **not** ↓ mortality.
- Inhaled or systemic corticosteroids: no consistent improvement in critical outcomes.
- **Inhaled nitric oxide** for MAS with pulmonary hypertension → ↓ death and ↓ ECMO.
- **ECMO** only when oxygenation is poor despite maximal ventilation: **0.35%** of >5100 MAS cases (California); mortality **5.3%**.

2. Neonatal Encephalopathy and Cerebral Palsy

- The old belief that intrapartum events cause most neurological disability drove the rise in cesareans from the 1970s. It was wrong.
- **Nelson and Ellenberg**: CP is multifactorial (genetic, physiological, environmental, obstetrical); **few cases are linked to peripartum events**.
- **ACOG Task Forces**: 2003, updated **2014** (more circumspect). The 2014 Task Force recommends a **multidimensional assessment** of each infant; no single strategy gives 100% certainty about cause.

Neonatal encephalopathy

- **Definition (2014 Task Force)**: neurological dysfunction in the earliest days of life in neonates **≥35 wk**, shown by **seizures or subnormal consciousness**, often with trouble starting/maintaining respiration and depressed tone and reflexes.
- Incidence **1–2 per 1000 term live births** (much more frequent preterm).
- Evaluation: maternal history, obstetrical antecedents, intrapartum factors, **placental pathology**, newborn course, labs, neuroimaging.

Grade	Features
Mild	Hyperalertness, irritability, jitteriness, hypertonia and hypotonia
Moderate	Lethargy, abnormal tone, occasional seizures
Severe	Coma, multiple seizures, recurrent apnea

- **Only spastic quadriplegic CP can result from acute peripartum ischemia.**
 - Hemiparetic/hemiplegic CP, spastic diplegia and ataxia are **unlikely** to come from an intrapartum event.
 - Purely **dyskinetic or ataxic** CP (especially with a learning disorder) is usually **genetic**.

Table 33-1. Findings Consistent with an Acute Peripartum or Intrapartum Event Leading to Hypoxic-Ischemic Encephalopathy

Category	Finding
Neonatal findings	Apgar score: <5 at 5 and 10 minutes
	Umbilical arterial acidemia: pH <7.0 and/or base deficit ≥12 mmol/L
	Neuroimaging evidence of acute brain injury: MR imaging or MRS consistent with HIE
	Multisystem involvement consistent with HIE
Type and timing of contributing factors	Sentinel hypoxic or ischemic event occurring immediately before or during delivery
	Fetal heart rate monitor patterns consistent with an acute peripartum or intrapartum event

HIE = hypoxic-ischemic encephalopathy; MR = magnetic resonance; MRS = magnetic resonance spectroscopy.

Caveats to the criteria

- **Apgar:** low 5- and 10-min scores ↑ risk, but most such infants do not get CP. **5-min Apgar ≥7 → peripartum HIE unlikely** to have caused CP.
- **Acid-base:** low pH/high base deficit ↑ likelihood of HIE on a continuum, but most acidemic neonates are neurologically normal. **Cord artery pH ≥7.2 → HIE very unlikely.**
- **Imaging: MR imaging/MRS is best;** cranial sonography and CT lack sensitivity at term.
 - **Normal imaging after the first 24 h effectively excludes a hypoxic-ischemic cause.**
 - MR at **24–96 h** is more sensitive for timing; MR at **7–21 days** best shows the full extent of injury.
- **Multisystem injury** (renal, GI, hepatic, cardiac, hematological) is consistent with HIE; its severity does not necessarily track neurological severity.
- **Sentinel events: uterine rupture, severe abruption, cord prolapse, amnionic fluid embolism.**
 - Martinez-Biarge (~58,000 deliveries, 192 sentinel events): **6% died** intrapartum/early neonatal, **10% developed encephalopathy.**
- Other risks for neonatal acidosis: prior or emergent cesarean, maternal age **≥35**, obesity, thick meconium, chorioamnionitis, general anesthesia.
- **FHR pattern on presentation vs later:**
 - Category 1 or 2 tracing with **5-min Apgar ≥7**, normal cord gases (± 1 SD) or both → **not** consistent with acute HIE.
 - A tracing **on admission** with persistently **minimal/absent variability, no accelerations, ≥60 min**, even without decelerations → suggests an **already compromised fetus**. If well-being cannot be established, evaluate method and timing of delivery.

Therapeutic hypothermia and adjuncts

- **Cooling started within 6 h of life, to 33.5°C for 72 h** → ↓ death and moderate–severe disability in term newborns.
 - Meta-analysis (>1500 neonates): better survival and neurodevelopment at **24 months**; protection extends into childhood.
 - MR: slower diffusion abnormalities, fewer infarcts.
 - **Longer or deeper cooling adds nothing.**
 - "Late" cooling (after 6 h) → some benefit, but less.
- Hospitals need plans for **timely transfer to a cooling center.**
- Adjuncts under study: **erythropoietin** (antiapoptotic, anti-inflammatory; meta-analysis shows some ↓ cognitive impairment and CP), allopurinol, xenon, melatonin, stem cells.

Cerebral palsy

- **Definition:** a group of **nonprogressive** disorders of movement or posture from abnormal development or damage of motor-control centers.
- Classified by dysfunction (**spastic, dyskinetic, ataxic**) and by limbs (**quadriplegia, diplegia, hemiplegia, monoplegia**).
 - **Spastic quadriplegia — the most common;** strongly linked to intellectual disability and seizures.
 - **Diplegia** — common in preterm/LBW neonates.
- Epilepsy and intellectual disability seldom stem from perinatal asphyxia **in the absence of CP**.

Incidence

- US prevalence **~3 per 1000 children** (all children, including preterm).
- Overall rate unchanged despite more cesareans, partly because more preterm infants survive (Fig 33-1).
- **23–27 wk: 105 per 1000** infants alive at 1 year (Sweden).
- **Term newborns make up about half of CP cases** in absolute numbers.

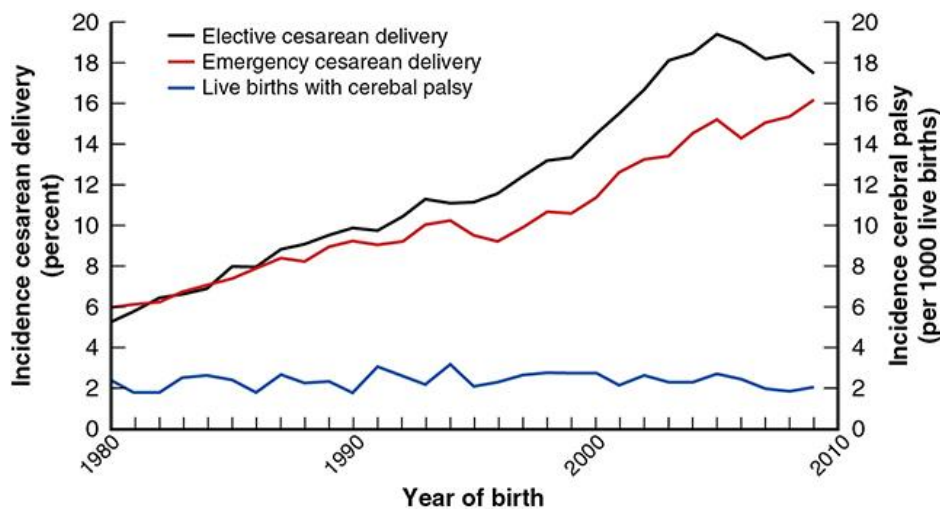


Figure 33-1. Elective and emergency cesarean rates climbed from ~6% to 16–19% (1980–2010), while CP stayed flat at ~2 per 1000 live births.

Nelson and Ellenberg (Collaborative Perinatal Project, ~54,000 pregnancies, followed to age 7) — main risk factors:

- (1) **Genetic abnormalities** (maternal intellectual disability, fetal malformations)
- (2) **Birthweight <2000 g**
- (3) **Birth <32 wk**
- (4) **Perinatal infection**
- Obstetrical complications were not strongly predictive; only **one fifth** had markers of perinatal asphyxia. Most causes unknown; no single intervention would prevent many cases.
- **Preterm birth is the single most important risk.** SGA also ↑ risk: **>90%** of CP in growth-restricted newborns is antepartum in origin.

Table 33-2. Perinatal Risk Factors Reported to Be Increased in Children with Cerebral Palsy

Risk Factors	Risk Ratio	95% CI
Hydramnios	6.9	1.0–49.3
Placental abruption	7.6	2.7–21.1
Interval between pregnancies <3 mo or >3 yr	3.7	1.0–4.4
Spontaneous preterm labor	3.4	1.7–6.7
Preterm delivery at 23–27 weeks	78.9	56.5–110
Breech or face presentation, transverse lie	3.8	1.6–9.1
Severe birth defect	5.6	8.1–30.0
Nonsevere birth defect	6.1	3.1–11.8
Time to cry >5 minutes	9.0	4.3–18.8
Obesity	1.2–2	1.1–2.8
Low placental weight	3.6	1.5–8.4
Placental infarction	2.5	1.2–5.3
Chorioamnionitis — clinical	2.4	1.5–3.8
Chorioamnionitis — histological	1.8	1.2–2.9
Others ^a	—	—

^aIncludes respiratory distress syndrome, meconium aspiration, emergent cesarean or operative vaginal delivery, hypoglycemia, gestational hypertension, hypotension, advanced maternal age, genetic factors, twins, thrombotic states, nighttime delivery, seizures, fetal-growth restriction, male gender, and nulliparity. CI = confidence interval. The printed "Severe birth defect" row (5.6; 95% CI 8.1–30.0) is internally inconsistent — the point estimate lies outside its CI. It is reproduced as printed.

Other causes and associations

- **Chorioamnionitis** carries substantively greater risk.
- **Neonatal arterial ischemic stroke** (may be linked to inherited fetal thrombophilias).
- Isolated congenital heart lesions → ↑ microcephaly (chronic fetal hypoxemia).
- Miscellaneous: fetal anemia, **TTTS**, intrauterine transfusions, fetal alcohol syndrome.
- **Only 1.6 CP cases per 10,000 deliveries are attributable solely to intrapartum hypoxia** (2003 Task Force). FGR and birth defects contribute more than asphyxial birth events.

Intrapartum FHR monitoring

- **Electronic FHR monitoring does not predict or reduce CP** or long-term neurological impairment (ACOG).
- No FHR pattern predicts CP; clinician response to abnormal patterns does not relate to outcome; computer analysis doesn't help.
- An abnormal tracing may reflect a **pre-existing** neurological abnormality.

Apgar scores

- 1- and 5-min Apgars are **poor predictors** of long-term neurological impairment.
- **5-min Apgar ≤3** → death and neurological sequelae rise substantially.
 - Sweden: **5%** later needed special schooling. Scotland (>750,000 term children): incidence **0.3%**, with a **dose-dependent** link to later educational support (physical/motor strongest, then visual, hearing, cognitive).
- Persistence of very low scores beyond 5 min strongly predicts morbidity and death:

Apgar	Mortality	Cerebral palsy
10-min 0–3	—	10%
15-min ≤2	53%	36%
20-min ≤2	60%	57%

Table 33-3. Comparison of Mortality and Morbidity in Norwegian Infants Weighing >2500 g According to 5-Minute Apgar Scores

Outcome	Apgar 0–3 (%)	Apgar 7–10 (%)	Relative Risk (95% CI)
Number	292	233,500	
Mortality rates			
Neonatal	16.5	0.05	386 (270–552)
Infant	19.2	0.3	76 (56–103)
1–8 yr	3	0.2	18 (8–39)
Morbidity rates			
Cerebral palsy	6.8	0.09	81 (48–128)
Mental retardation	1.3	0.1	9 (3–29)
Other neurological	4.2	0.5	9 (5–17)
Non-neurological	3.4	2.0	2 (0.8–5.5)

CI = confidence interval.

Umbilical cord blood gases

- **Metabolic acidosis = cord artery pH <7.0 and base deficit ≥12 mmol/L** → risk factor for encephalopathy and CP; risk grows as acidosis worsens.
- Alone, cord gases are **not accurate** predictors of long-term sequelae.
- **pH <7.0 = threshold for clinically significant acidemia.**
- Casey 2001: pH ≤6.8 → neonatal mortality ↑ 1400-fold; pH ≤7.0 plus 5-min Apgar 0–3 → ↑ 3200-fold.
- Oxford: adverse neurological outcome **0.36% with pH <7.1, 3% with pH <7.0.**
- **Cord lactate** may also help predict outcome.

Nucleated RBCs and lymphocytes

- Both enter the circulation in response to hypoxia or hemorrhage. Proposed as markers of hypoxia, but data are inconsistent.

Neuroimaging

- Findings vary with GA, insult severity/duration and reperfusion. **Precise timing of injury by imaging is not realistic** — only an imprecise window.
- Encephalopathy grade does **not** correlate with MR findings.

Modality (term newborn)	Day of birth	With injury
Sonography	Usually normal	Thalami/basal ganglia increasingly echogenic from ~24 h , progress over 2–3 days, persist 5–7 days
CT	Usually normal	Thalami/basal ganglia ↓ density from ~24 h , persist 5–7 days
MR imaging	Some abnormalities detectable	Restricted diffusion within 24 h, peaks ~5 days, gone within 2 wk ; T1/T2 changes from <24 h to several days

- **Basal ganglia lesions on MR** accurately predicted motor impairment at 2 yr (175 term neonates).

Older children with CP

- Wu (273 children born >36 wk, CT/MR): **1/3 normal**; focal arterial infarction **22%**; brain malformations **14%**; periventricular white-matter injury **12%**.
- MR abnormal in **88%** (Bax; ~half near term) and in **84% of spastic quadriplegia**.
- MR is preferred over CT (radiation).
- Hemiplegic CP at term (Wiklund): ~75% abnormal CT; **>1/2 suggested prenatal injury**, ~20% perinatal.

Intellectual disability, seizures and autism

- When either occurs **without CP**, it is seldom from perinatal hypoxia.
- **Severe intellectual disability: 3 per 1000**; commonest causes are chromosomal, gene mutations and other malformations; also preterm birth.
- **Seizure disorders**: major predictors are fetal malformations (cerebral and noncerebral), family history of seizures, and neonatal seizures. Encephalopathy causes a small share; seizures correlate best with **encephalopathy severity**.
- **Autism spectrum disorder: 15 per 1000** 8-year-olds; not convincingly linked to peripartum events.

3. Neonatal Abstinence Syndrome

- Drug withdrawal, most often after in-utero **opioid** exposure; also alcohol or benzodiazepines.
- Features: **hypertonia, autonomic instability, irritability, poor suck, seizures**.
- Incidence ↑ **6–7-fold** over the past decade; **4% of NICU days** in 2013.
- Care: close observation + usually pharmacotherapy — **morphine, methadone**; also phenobarbital, benzodiazepines, clonidine.
- **Buprenorphine vs morphine** → **shorter treatment and hospital stay**. No consensus on the best regimen.

4. Hematological Disorders

Anemia

- After **35 wk**, mean cord Hb **~17 g/dL**; **<14 g/dL is abnormal**.
- **Delayed cord clamping**:
 - Hb **+1.2 g/dL** at birth; improves iron stores for several months.
 - Anemia at 6 months not different.
 - ↑ **hyperbilirubinemia needing phototherapy** (no longer stay).
 - **ACOG: delay 30–60 seconds in all healthy newborns**.
- Causes:
 - **Antenatal (extension of fetal anemia)**: immune-mediated hemolysis, aneuploidy, genetic hematological defects, fetomaternal hemorrhage, fetal infection, placental lesions (chorioangioma, twin anastomoses).
 - **At delivery (acute, with hypovolemia)**: laceration of placenta, fetal vessel or cord; intracranial/extracranial or intra-abdominal organ hemorrhage.

Polycythemia

- Associations: chronic or acute fetal hypoxia, **TAPS**, maternal diabetes or hypertension, placental and fetal-growth restriction, aneuploidy, delayed cord clamping.
- **Hct >65** → **viscosity rises markedly**. No routine screening in asymptomatic newborns.
- Signs: hypotonia, **plethora**, respiratory distress with cyanosis, jitteriness, **hypoglycemia**; hyperbilirubinemia follows (short-lived macrocytic fetal RBCs).

- **Isovolemic partial exchange transfusion (PET) with saline** ↓ Hct and symptoms but is **not neuroprotective**; usually reserved for acute symptoms or hypoglycemia.

Hyperbilirubinemia

- Fetal bilirubin is partly cleared across the placenta. After birth, levels rise for **3–4 days to up to 10 mg/dL**, then fall rapidly.
- **~15%** of term newborns become jaundiced.
- ↑ Burden: polycythemia, hemolysis (**ABO incompatibility, G6PD deficiency**), extravasated blood (**cephalohematoma**). ↓ Clearance: prematurity.

Form	Features
Acute bilirubin encephalopathy	First days of life: hypotonia, poor feeding, lethargy, abnormal auditory-evoked responses. Prompt treatment usually halts it.
Kernicterus (chronic)	Bilirubin staining of basal ganglia and hippocampus → neuronal degeneration. Survivors: spasticity, incoordination, intellectual deficits. Ongoing hemolysis is a risk factor.

- Prevention: **effective breastfeeding** (avoid dehydration); check for jaundice with vital signs; **universal predischage bilirubin screening** (serum or, more often, **transcutaneous**).
- **Phototherapy ("bili-lights") 460–490 nm** → bilirubin photo-oxidation → renal clearance. Filtered sunlight (UV removed) in low-resource settings.
- Kernicterus hospitalizations: **5 per 100,000 (1988) → 0.4–2.7 per 100,000**; persistence partly linked to **early discharge**.

Vitamin K deficiency bleeding (VKDB)

- Causes of deficiency: poor antenatal liver stores, **sterile gut, low vitamin K in breast milk**.

Onset	Timing	Notes
Early	<24 h	e.g., maternal vitamin K deficiency
Classic	Days 2–7	No prophylaxis at delivery
Late	2–12 wk	Exclusively breastfed infants

- **Prophylaxis: single IM dose of vitamin K1 (phytonadione) 0.5–1 mg** (AAP/ACOG).
 - Oral is less effective; maternal vitamin K barely reaches the fetus.
 - **No link to childhood leukemia.**
- "Hemorrhagic disease of the newborn" is broader: also congenital factor deficiency, DIC, birth trauma, thrombocytopenia.

Thrombocytopenia

- **Early onset ≤72 h**, late onset after.
- Causes: **immune** (alloimmune thrombocytopenia, maternal ITP, neonatal lupus — antiplatelet IgG crosses to the fetus), **infection** (parvovirus B19, CMV, toxoplasmosis; sepsis → consumption), maternal **antiretroviral therapy**, inherited platelet disorders, congenital syndromes.

5. Newborn Injuries

Incidence

- Birth injury **~2% of deliveries** (varies with how major and minor injuries are lumped).
- Washington State: **5.3 → 4.5 per 1000** live births (2004–2013), a **14%** fall.
- **Major injury highest with forceps or vacuum**, much lower with all cesarean categories.

Cranial injuries

Table 33-4. Major Types of Cranial Hemorrhage

Type	Collection Site	Associations in Term Neonates	Clinical Outcomes
Intracranial			
Epidural	Between periosteum and dura mater	Difficult or prolonged birth; idiopathic	Usually good
Subdural	Between dura and arachnoid mater	Difficult or prolonged birth; idiopathic	Usually good
Subarachnoid	Between arachnoid and pia mater	Difficult or prolonged birth; idiopathic	Usually good
Intraventricular	Ventricles (lateral, 3rd, or 4th)	Birth trauma, hypoxia, or coagulation dysfunction	Variable; worse with high IVH grade and thalamic source
Intracerebral	Cerebral parenchyma	Coagulation dysfunction, vascular anomalies, intrapartum factors	Variable; affected by size, location, or comorbid hydrocephalus
Intracerebellar	Cerebellar parenchyma	Abnormal FHR, OVD, resuscitation required	Variable; affected by size and comorbid lesions
Extracranial			
Cephalohematoma	Subperiosteal	OVD, especially VAD; dystocia	Good; risk of anemia or hyperbilirubinemia
Subgaleal	Between galea aponeurotica and periosteum	OVD, especially VAD; dystocia	Variable; worse with associated acidosis, hypovolemia, and comorbid lesions

OVD = operative vaginal delivery; VAD = vacuum-assisted delivery. (The epidural, subdural and subarachnoid rows share one bracketed "Associations" entry in the printed table.)

Intracranial hemorrhage

- **Term: subdural and intracerebral** are most common. **Preterm: intraventricular, subarachnoid, cerebellar** (BP instability, hypoxia, ischemia).
- At term, **birth trauma** is the most frequent cause; others: fetal coagulopathy, vascular anomaly, tumor, genetic mutation, infection. Some follow an uneventful vaginal birth with no cause found.
- Presentation: **seizures, apnea or tachypnea, poor feeding, temperature instability.**
- Imaging: sonography or CT; **MR is often preferred.** Treatment mostly supportive; surgery for acute hydrocephalus, ↑ ICP or mass effect.
- **Subdural:** prolonged/difficult delivery → molding, dural stretch, **tearing of bridging veins.** Also seen asymptotically after uneventful vaginal or cesarean birth; **good prognosis if asymptomatic.**
- Subarachnoid (less common) and epidural (rare): risks and recovery like subdural.
- **IVH at term:** germinal matrix has largely regressed, so the source is the **choroid plexus, thalamus or residual germinal matrix.** $\frac{1}{3}$ – $\frac{2}{3}$ have no antecedent. Worse with high grade or **thalamic** hemorrhage.
- **Intracerebral (hemorrhagic stroke):** coagulation defects, vascular anomalies, abnormal FHR, low Apgars; trauma less often.
- **Intracerebellar:** abnormal FHR, operative vaginal delivery, emergent cesarean, delivery-room resuscitation.

Extracranial hemorrhage

- Scalp layers from outside in: **skin → subcutaneous tissue → galea aponeurotica → subgaleal space → periosteum → calvarium.** **Emissary veins** cross the subgaleal space, linking dural sinuses to scalp veins.

Lesion	Layer	Crosses sutures?	Key features
Caput succedaneum	Scalp soft-tissue edema	Yes	Pressure of head on cervix; maximal at birth , gone within hours–days
Cephalohematoma	Subperiosteal (emissary/diploic veins)	No	Incidence ~1%; dystocia, OVD (vacuum > forceps or cesarean). Appears hours after birth, grows, persists weeks–months → anemia, hyperbilirubinemia. Rarely infection or calcification. Good outcome.
Subgaleal hemorrhage	Between galea and periosteum (emissary veins)	Yes (spans occipital, parietal, frontal)	Rare; 0.05% spontaneous, 10× with vacuum . Can spread from neck to orbits → hypovolemic shock; mortality ~15% . Seizures, enlarging head circumference, respiratory distress.

- Diagnosis starts with **head ultrasound** (may suffice for cephalohematoma); CT/MR for subgaleal hemorrhage to exclude other injury.

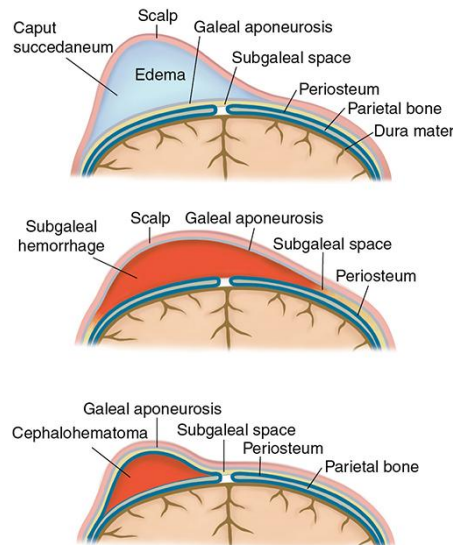


Figure 33-2. Caput = edema above the galea; subgaleal hemorrhage = blood between galea and periosteum (spreads widely); cephalohematoma = blood under the periosteum (limited by sutures).

Memory hook: "**C**ephalohematoma **C**an't cross sutures; **S**ubgaleal **S**preads and causes **S**hock."

Skull fractures

- Rare (~1 per 100,000 births) but worrying because of associated **intracranial hemorrhage**.
- Associated with operative vaginal delivery, but also spontaneous or cesarean birth: compression against the pelvis, surgeon's hand lifting the head, or **upward vaginal pressure by an assistant** to disimpact the head.
- Diagnose with radiographs or CT; expectant or surgical management. Prognosis depends on comorbid intra-/extracranial injury.



Figure 33-3. Depressed skull fracture after cesarean for a deeply impacted head disimpacted by upward vaginal hand pressure.

Spinal cord injury

- Rare overstretch with hemorrhage/edema; **cervical spine** most often.
- Excess traction, rotation or hyperextension during **shoulder dystocia maneuvers, breech vaginal delivery, operative vaginal delivery**.
- **Upper cervical injury is often fatal**; survivors may be quadriplegic. Treatment: ventilatory support with corticosteroids or therapeutic hypothermia.

Peripheral nerve injuries

Brachial plexopathy (roots C5–C8 and T1)

- ~1 per 1000 term births: 1.3/1000 vaginal vs 0.2/1000 cesarean.
- Risks: **shoulder dystocia**, labor dystocia, breech delivery.
- Hemorrhage/edema → temporary dysfunction, good recovery. **Avulsion** → **poor prognosis**.

Palsy	Roots	Findings
Erb	C5–6 (upper trunk)	Paralysis of deltoid, infraspinatus and forearm flexors. Arm straight and internally rotated , elbow extended, wrist and fingers flexed. Finger function retained. Typical breech injury.
Klumpke	C8–T1 (lower plexus)	Flaccid hand
Total plexus	C5–T1	Flaccid arm and hand

Pearl: the Erb posture (adducted, internally rotated arm, extended elbow, flexed wrist) is classically called the "waiter's tip" position.

- Severe injury: **phrenic nerve** → ipsilateral diaphragm paralysis; sympathetic fibers → **Horner syndrome (ptosis, miosis, anhidrosis)**.
- Clavicle or humerus fracture may coexist → chest radiograph. MR for suspected avulsion.
- Breech injuries are usually Erb type; more extensive, persistent lesions follow difficult cephalic (forceps/vacuum) deliveries.
- **ACOG Task Force:** shoulder dystocia **cannot be accurately predicted**; in most cases axons survive and prognosis is good.
- **Recovery (Lindqvist): C5–6 86% complete; C5–7 38%; global C5–T1 → always permanent disability.**
- Management: **early physical therapy** (prevent contracture); surgical exploration/nerve reconstruction for severe injury at **3–6 months**.

Facial paralysis

- Nerve injured at the **stylomastoid foramen**. **0.3 per 1000 term births**.
- **¾ after spontaneous birth** (pressure from bony pelvis), **¼ with forceps** (blade pressure); can follow cesarean.
- **Spontaneous recovery within days is the rule**; permanent paralysis described.



Figure 33-4. Left facial nerve palsy: eye fails to close, labionasal fold lost, mouth pulls to the unaffected side when crying.

Fractures

- After any difficult delivery, **palpate clavicles and long bones**; crepitus or irregularity → radiograph.
- **Clavicle**: commonest; **~5 per 1000** live births; unpredictable and **unavoidable**; **10× higher with vaginal birth**. Vaginal: shoulder dystocia; cesarean: higher birthweight.
- **Humerus** (rare): breech delivery, **posterior arm delivery** in shoulder dystocia.
- **Femur** (rare): breech — now mostly at **cesarean**.
- Prevention: splint the bone along its long axis; avoid perpendicular forces during extraction.
- **Rib fractures** rare at term; from chest compression in difficult birth/shoulder dystocia (none from CPR in one series).

Soft tissue injury

- **Intra-abdominal hemorrhage** (rare): **liver, spleen, adrenals** (disproportionately large).
- Risks: difficult (especially breech) delivery, cardiac resuscitation, macrosomia, organ enlargement.
- Retroperitoneal bleeding → bruising of flanks, umbilicus, groin, upper thigh or scrotum; pallor, distension, falling Hct. Treat with resuscitation and bleeding control.

Congenital deformity injuries

- **Congenital torticollis**: one sternocleidomastoid fails to elongate; head turns toward the affected side. Linked to breech, OVD, difficult delivery, but many are uncomplicated births. **Early physical therapy**.
- **Deformations**: normal structures deformed by restrictive intrauterine forces — **chronic oligohydramnios, multifetal gestation, abnormal/small uterine cavity**.
 - **Talipes equinovarus**; **oligohydramnios sequence** (pulmonary hypoplasia; head, face, skin, limb deformations). Both can also be genetic or with malformations.
- **Congenital hip dislocation**: strongly linked to **in-utero breech**. Examine all newborns; observe or brace.

Quick-Recall: Differentials and Distinctions

Topic	Key distinction
TTN vs RDS vs MAS (term)	TTN: retained lung fluid, cesarean without labor, <39 wk, resolves ≤ 48 h, diagnosis of exclusion. RDS: surfactant deficiency (male, white, diabetes, SP-B/SP-C/ABCA3). MAS: postterm/FGR, thick meconium, pulmonary hypertension
MAS prevention	Nothing works: no perineal suction, no routine intubation, no amnioinfusion
MAS treatment	Surfactant ($\downarrow$ ECMO, not death); iNO ($\downarrow$ death and ECMO); ECMO last resort
CP type from intrapartum ischemia	Spastic quadriplegia only ; dyskinetic/ataxic = genetic
HIE unlikely if...	5-min Apgar ≥ 7 ; cord pH ≥ 7.2 ; normal imaging after 24 h
HIE consistent if...	Apgar < 5 at 5 and 10 min; pH < 7.0 and/or BD ≥ 12 ; MR/MRS changes; multisystem injury; sentinel event; consistent FHR
Cooling	Start ≤ 6 h; 33.5°C ; 72 h; longer/deeper no better
Caput vs cephalohematoma vs subgaleal	Edema, crosses sutures, max at birth / subperiosteal, respects sutures, appears later / below galea, spreads, shock (15% mortality)
Term vs preterm ICH	Term: subdural, intracerebral (trauma). Preterm: IVH, subarachnoid, cerebellar
Erb vs Klumpke	C5–6: waiter's tip, fingers work / C8–T1: flaccid hand $\pm$ Horner
VKDB timing	Early < 24 h; classic 2–7 d; late 2–12 wk (breastfed)
Kernicterus site	Basal ganglia and hippocampus

Key Numbers to Memorize

- TTN resolves within **48 h**; severe MAS **4/1000**; MAS needing ECMO **0.35%** (mortality **5.3%**)
- Neonatal encephalopathy: **≥ 35 wk**; **1–2/1000** term births
- HIE criteria: Apgar **< 5 at 5 and 10 min**; cord pH **< 7.0** and/or BD **≥ 12 mmol/L**
- Cooling: **within 6 h**, **33.5°C** , **72 h**
- CP prevalence **3/1000**; 23–27 wk **105/1000**; intrapartum hypoxia alone **1.6/10,000** deliveries; cesarean rate **$\uparrow 6\times$** with no change in CP
- Apgar 0–3 at 5 min $\rightarrow$ neonatal death RR **386**, CP RR **81**; 10-min 0–3 $\rightarrow$ CP **10%**; 20-min $\leq 2 \rightarrow$ death **60%**, CP **57%**
- Cord pH **≤ 6.8** $\rightarrow$ neonatal death **$\uparrow 1400\times$** ; pH ≤ 7.0 + Apgar 0–3 $\rightarrow$ **$\uparrow 3200\times$**
- MR diffusion restriction peaks **~ 5 days**, gone by **2 wk**; best extent at **7–21 days**
- Cord Hb **~ 17 g/dL** (abnormal **< 14**); delayed clamping **30–60 s** (+1.2 g/dL); polycythemia **Hct > 65**
- Bilirubin peaks day **3–4**, up to **10 mg/dL**; jaundice **15%**; phototherapy **460–490 nm**
- Vitamin K1 **0.5–1 mg IM** once
- Birth injury **$\sim 2\%$** ; cephalohematoma **1%**; subgaleal **0.05%** ($\times 10$ with vacuum), mortality **15%**; skull fracture **1/100,000**
- Brachial plexus **1/1000** (vaginal 1.3 vs cesarean 0.2); C5–6 recovery **86%**; facial palsy **0.3/1000**; clavicle fracture **5/1000** ($10\times$ vaginal)